Solutions for Homework 3-4.

1. a) $\mathrm{Var}\, X = \frac{105560}{6561} = 16.085$

b) $\mathrm{Var}\, Y = \frac{65}{4} = 16.25$

This question may be of interest: given the possible values of a r.v., like $x_1, \cdots, x_n$, what distribution on $\{x_1, \cdots, x_n\}$ will give maximum variance (the minimum variance is 0)?

2. The probability is

$$\binom{s+t-1}{s} p^{s+1} (1-p)^{s+t-1} \quad (\text{Why?})$$

3. Flip the coin twice.

$$\mathbb{P}(HT) = p(1-p) = \mathbb{P}(TH).$$

Therefore, we flip the coin twice as many times as needed to obtain different outcome (HT or TH). Then the probability of HT is $\frac{1}{2}$.

4. Flip the coin twice till one of the outcomes is a head. Then $\mathbb{P}(HH) = \frac{1}{3}$.

5. From Ex 3. we know how to have an event with prob. $\frac{1}{2}$. Now flip the coin 5 times till we get one of the following outcomes:

HTHTHT

HTHTTH

HTTHHT

HTTHTH

THHTHT .

Then the probability of HTHTHT is 1/5.

6. a) If you spend $x, the the expected value of your money at the end of the week is

$$\frac{x}{2} \cdot \frac{4+1}{2} + 1000 - x = 1000 + 0.25x ,$$

which is maximal when $x = 1000$.

b) If you spend $x then the expected amount of the commodity by the end of the week is

$$\frac{x}{2} + \frac{1}{2} \cdot (1000 - x) + \frac{1}{2}(1000 - x) \cdot \frac{1}{4} = \frac{5}{8} \cdot 1000 - \frac{x}{8} ,$$

which is maximal of $x = 0$.

7. The probability that we need one test is 0.9^{10}.
The probability that we need eleven tests is $1 - 0.9^{10}$.
Therefore the expected number of tests is

$$0.9^{10} \cdot 1 + (1 - 0.9^{10}) \cdot 11 = 7.51 ,$$

instead of 100 tests we need only 75.1 tests on average.